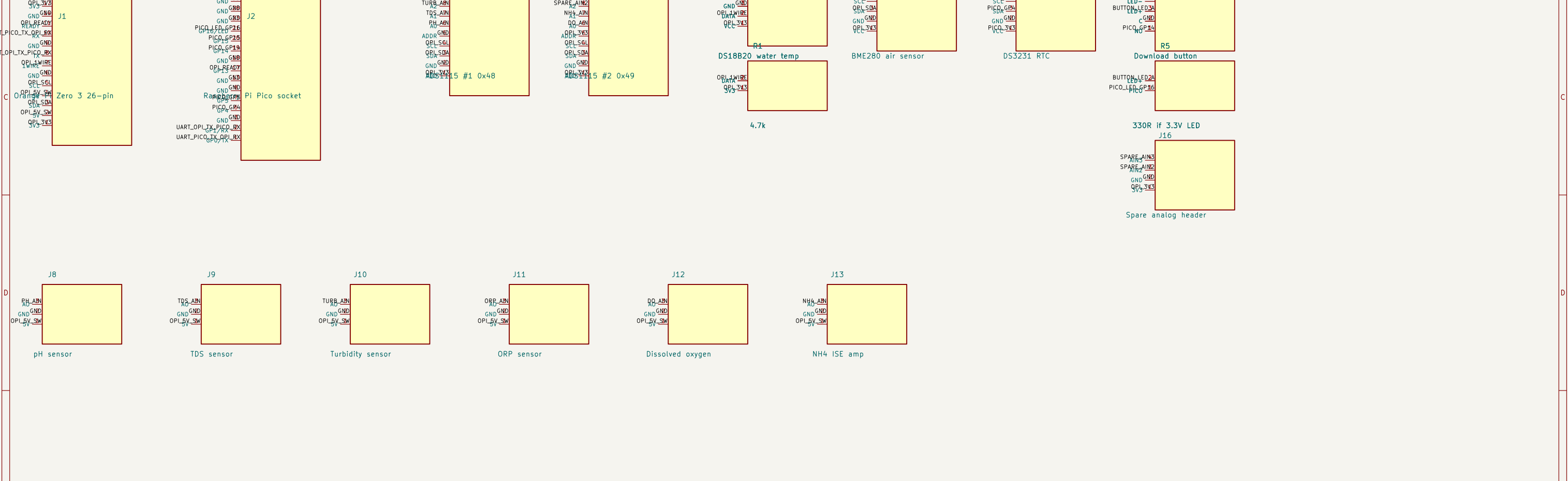


The schematic diagram illustrates the power supply section of the Orange Pi 3B+, showing the flow of power from the USB-C input to the various components of the board. The diagram is divided into several functional blocks:

- USB-C Input:** The input is connected to the USB-C port (J14) and the USB-C PD input (USB_C_PD).
- VBUS TVS:** A transient voltage suppressor (D1) is used to protect the VBUS line from voltage spikes.
- PD Sink:** The PD sink (U1) is responsible for negotiating the power profile with the host. It is connected to the USB-C PD input and the VBUS line.
- 3A-5A Polyfuse:** A polyfuse (F1) is used to limit the current in the VBUS line, providing overcurrent protection.
- Buck Converter:** The buck converter (U2) is used to step down the VBUS voltage to the 5V rail. It is connected to the VBUS line and the 5V rail.
- 5V Rail:** The 5V rail is the output of the buck converter and is used to power the various components of the board. It is connected to the 5V pin of the USB-C port and the 5V pin of the power jack.
- Power Management ICs:** The power management ICs (U1, U2) are responsible for managing the power flow and ensuring that the components of the board are powered correctly.
- Capacitors:** Various capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100) are used to filter the power and ensure that the components of the board are powered correctly.
- Diodes:** Various diodes (D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24, D25, D26, D27, D28, D29, D30, D31, D32, D33, D34, D35, D36, D37, D38, D39, D40, D41, D42, D43, D44, D45, D46, D47, D48, D49, D50, D51, D52, D53, D54, D55, D56, D57, D58, D59, D60, D61, D62, D63, D64, D65, D66, D67, D68, D69, D70, D71, D72, D73, D74, D75, D76, D77, D78, D79, D80, D81, D82, D83, D84, D85, D86, D87, D88, D89, D90, D91, D92, D93, D94, D95, D96, D97, D98, D99, D100) are used to protect the components of the board from reverse polarity and overvoltage.



ES1 datasheet: short SW loop, thermal copper, tight input caps.

Check pinouts against supplier datasheets before PCB order.

							Review selected IC pinouts and buck layout before PCB manufacture. USB-C PD, buck regulator, Pico controller, Orange Pi header, ADCs, sensors, and RTC. Water Logger Sheet: / File: zero3-water-logger-carrier.kicad_sch Title: Orange Pi Zero 3 Water Logger Carrier <table border="1"> <tr> <td>Size: A3</td> <td>Date: 2026-05-25</td> <td>Rev: R1 schematic</td> </tr> <tr> <td colspan="2">KiCad E.D.A. 10.0.3</td> <td>Id: 1/1</td> </tr> </table>	Size: A3	Date: 2026-05-25	Rev: R1 schematic	KiCad E.D.A. 10.0.3		Id: 1/1	
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